

Case Study: AI-Assisted Mentor–Startup Matching

Situation

Matching mentors (industry experts) with mentees (early-stage startups) for an online accelerator can be time-consuming, not just the matching itself but also identifying the gaps.

In addition, the process must account for mentor capacity, schedule alignment, and overall availability.

While AI tools exist for survey creation and matching, maintaining strict compliance with data privacy regulations was an essential requirement for this project.

Solution

The solution to this was to

- Create separate surveys for mentors and mentees using a compliant survey tool to gather matching criteria and inform wider accelerator programming (e.g., group workshop topics).
- Fully anonymise all survey responses prior to matching using an AI tool.
- **Use an AI tool** (Claude) to evaluate candidates using weighted criteria:
 - **50%:** Challenge vs. support fit (evaluating core blockers, desired support, and mentor expertise)
 - **25%:** Availability alignment
 - **12.5%:** Startup stage
 - **12.5%:** Business model alignment
- **Review** the initial AI-generated shortlists and manually adjust matches based on our experience of the startups and mentors, and identify and reach out to more potential mentors to fill the gaps.

The prompt I used:

"We run a US edtech accelerator and need to match founders (mentees) with mentors, using the attached anonymized survey responses.

Find the 3 best matches per mentee, weighted: 50% challenge/support fit (mentee's Top 3 Challenges, Most Urgent Challenge & Why, What's Blocking Them, Support Types Wanted, Desired Mentor Expertise vs. mentor's Top Expertise Areas and Mentoring Style), 25% availability, 12.5% Stage, 12.5% Business Model.

Hard exclusion: never surface a mentor against a mentee in a segment they've listed under "prefer not to be matched with." Cap each mentor at 5 total top-3 appearances (any rank). If two mentors are within 5% on score, prefer the one with fewer appearances so far.

Output: Excel table — mentee ID, mentor ID, rank (1-3), score, one-line rationale. Add a sheet for the bottom 10% of mentees by top-match score, flagged as gap cases. Add a sheet on expertise gaps with key mentor personas to recruit."

This can now be used as a reusable skill/template.

Impact & Key Takeaways

- Reduced manual matching time significantly compared to the fully manual process used in the previous cohort, using a process that's now reusable for future cohorts.
- The automated gap analysis sheet identified 5 recruitment personas, giving specific, actionable criteria for recruiting additional mentors, instead of relying on gut feel about who was "missing."
- Delivered high-precision matching without exposing sensitive personal data, keeping the process compliant with data privacy requirements throughout.

Future Enhancements

- Add a survey question asking mentors if they are willing to host general masterclasses for the entire cohort.
- Prompt Claude to analyse emerging industry trends to proactively align mentor recruitment with future-proof skill requirements.
- Integrate a lightweight communication style assessment to match mentors and mentees based on the mentoring style that meets their needs.